Santiago Orozco Holguín

Software Engineer

Medellín, Colombia + Remote · santiago@orozcoh.com · www.linkedin.com/in/orozcoh · github.com/orozcoh · santiago.orozcoh.com · cv.orozcoh.com

SUMMARY

Software Engineer with an Electronics Engineering background and experience designing, building, deploying, and maintaining web applications from concept to production. Passionate about turning ideas into products through hands-on engineering, with work spanning Progressive Web Applications (PWAs), distributed systems, cloud services, authentication, cryptography, and AI-assisted development workflows. Seeking to contribute to a collaborative engineering team while continuing to grow through real-world product development.

EXPERIENCE

Independent Software Engineer — Self-employed

Jun 2023 – Present · Remote · Self-employed

- Designed, developed, deployed, and maintained multiple production-ready web applications from concept to deployment using modern web technologies.
- Built Progressive Web Applications (PWAs) with a focus on mobile-first experiences, local-first principles, and maintainable software architecture.
- Designed cloud-native solutions using Cloudflare Workers, Cloudflare KV, REST APIs, and edge infrastructure.
- Developed software products exploring authentication, cryptography, passkeys, blockchain technologies, AI-assisted workflows, and distributed systems.
- Independently managed the complete application lifecycle, including architecture, implementation, domain management, DNS configuration, deployment, monitoring, and ongoing maintenance.
- Used AI-assisted development workflows to accelerate research, prototype solutions, validate implementation approaches, and explore unfamiliar technologies while maintaining ownership of engineering decisions.
- Built interactive software experiences integrating TouchDesigner licensing, telemetry collection, and cloud-based license management.

Software Engineering Intern — TSO Mobile Colombia

Oct 2022 – May 2023 · Medellín, Colombia · Internship

- Migrated a JavaScript UI component library to TypeScript, improving maintainability and type safety.
- Packaged reusable UI components into an internal npm library for use across multiple projects.
- Documented reusable components using Storybook, improving discoverability and developer experience.
- Collaborated with the engineering team to modernize frontend development workflows and reusable component architecture.

PROJECTS

career-os

Resume infrastructure as software

Single source of truth for professional identity. Structured YAML data compiles into ATS-friendly PDF, HTML, DOCX, and Markdown resumes plus a static website.

- Tag-based template system generates four targeted resumes from one dataset with zero duplication.
- Headless-Chromium print pipeline produces machine-readable, ATS-parseable PDFs from the same HTML the site serves.

Technologies: Astro, TypeScript, Tailwind CSS, Zod, Playwright, pnpm

Source

Shader Sketches

Generative art in the browser

A collection of interactive GLSL sketches exploring noise fields, particle systems, and audio-reactive visuals, packaged as embeddable web components.

- 60fps particle systems with 100k+ particles via GPU instancing.
- Featured in an online generative-art showcase with 20k visitors.

Technologies: WebGL, GLSL, TypeScript, Vite

Live · Source

SKILLS

Languages: TypeScript, JavaScript, Python, Go, SQL

Frontend: React, Astro, Tailwind CSS, Accessibility (WCAG)

Backend & Infrastructure: Node.js, PostgreSQL, Event-driven architecture, Cloudflare Workers, Docker

Creative Technology: WebGL / GLSL, Canvas 2D, Generative art, Web Audio API

Practices: Testing (unit, contract, e2e), CI/CD, Technical writing & RFCs

EDUCATION

Bachelor's Degree Electronics Engineering — Universidad de Antioquia

Aug 2014 – Jun 2023 · Medellín, Colombia

- Developed a strong foundation in software engineering, embedded systems, electronics, mathematics, and computer architecture.
- Completed an engineering internship focused on modern web development, TypeScript migration, and reusable component libraries.
- Transitioned academic interests toward software engineering through independent product development and modern web technologies.

LANGUAGES

- Spanish — Native
- English — Fluent